

Week 9 Workshop

Consider AR(1)-ARCH(1) model

$$\begin{aligned}r_t &= c + \phi_1 r_{t-1} + \varepsilon_t, \\ \varepsilon_t &= \sigma_t u_t, \quad u_t \sim i.i.d \, N(0, 1) \\ \sigma_t^2 &= a_0 + a_1 \varepsilon_{t-1}^2,\end{aligned}$$

with $a_0 > 0$ and $0 \leq a_1 < 1$. Derive

1. $E(\varepsilon_t)$ using the law of iterated expectations;
2. $V(\varepsilon_t)$ using the Law of Total Variance;
3. $E[(\varepsilon_t - E(\varepsilon_t))(\varepsilon_{t-k} - E(\varepsilon_{t-k}))]$ for all $k \geq 1$;

What do the above properties imply about ε_t ?